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On the Closure of Compositional Hierarchies in Digital Symmetries

A technical study of compositionality, closure, and symmetry in digital systems

Author

This monograph develops a rigorous framework connecting category theory, Boolean function theory, and applications in hardware, software, quantum computation, cryptography, and machine learning. It emphasizes executable artifacts, reproducibility, and multi-resolution error metrics.

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Front matter and technical monograph

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Prepared for readers with a background in discrete mathematics, basic category theory,
and digital logic